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The Changing Course of Storage in an AI World

OCP Global 2026

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The Evolution of a Storage Server

Storage servers have evolved over the decades. Since the development of NVMe, it has evolved even faster. Every generation of PCIe has resulted in big changes in the balance of compute/capacity/network.

Today storage architects are struggling with how to design storage servers for AI. What changes will be needed?

First, let's start with a definition for a Storage Server:

- Contains compute to handle requests. May be f/b/o or a combination.
- Contains 2 or more media devices (HDD or SSDs) for base storage.
- Provides resiliency in the event of the loss of one of the media devices.
- Typically intended to handle capacities of < 100 TB.

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Anatomy of Past Storage Servers in PCIe Gen 4 Era

What was the state of the art architecture of storage servers in PCIe Gen 4 era?

100 Gb NIC
- Second NIC may be added for failover
- Traffic from unattached CPU may be bottlenecked due to NUMA
- Limits speeds to < 12 GB/s per host

1400 W PSU
- 24 x 25W SSDs require 600W just for SSDs
- CPUs needed up to 250 W each

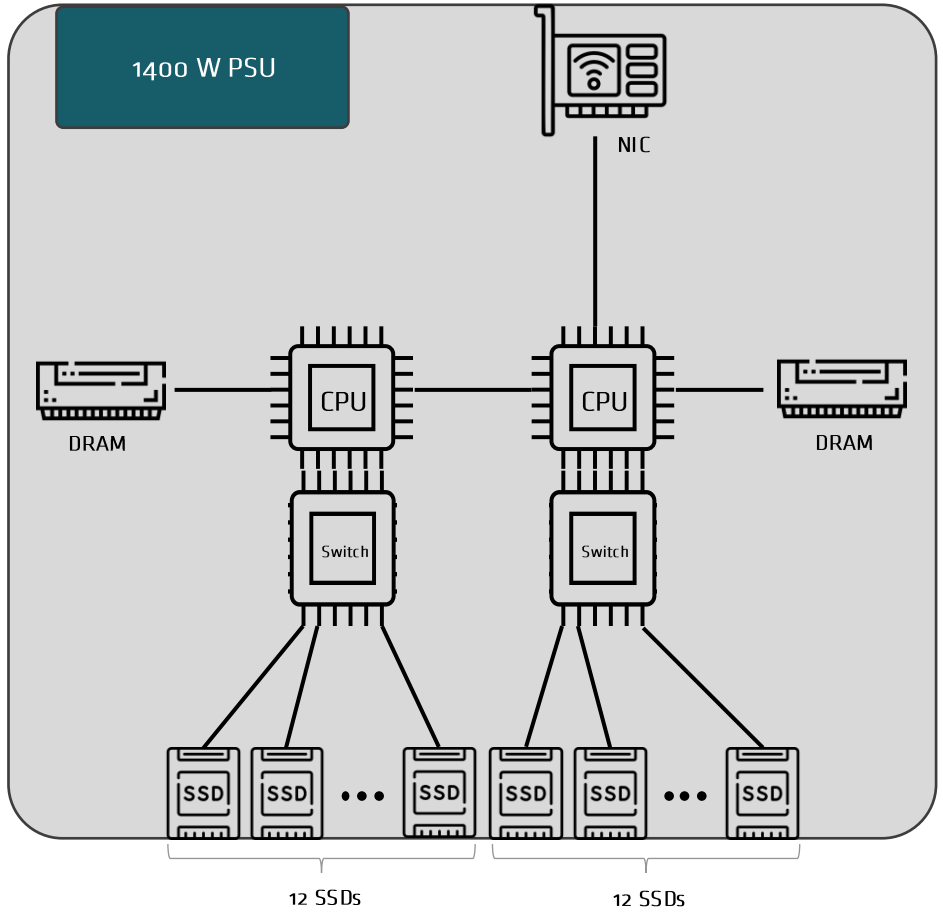
2 x 24 core (x86 CPUs)
- Performance often CPU limited
- Core counts vary
- Some companies begin to switch to single CPU AMD CPUs

128 – 384 GB DRAM
- Capacity needed varies depending on software
- DRAM may be used for caching

PCIe Switches
- Used for HA
- Also to connect directly to 12 SSDs x4 (or 24 x2)

24 x 8 TB SSDs (U.2 form factor)
- U.2 form factor limits to 24 SSDs in 2U chassis
- Often supports dual-port for HA
- Drives limited to 4 and 8 TB
- Up to 192 TB of RAW SSD capacity per server

Standard 2U, 24-SSD Storage Server



Issues

- Performance often limited by compute. Forces architects to use relatively expensive high-core count CPUs
- Network limit of 100Gb acceptable with low overall system performance
- System design often requires expensive HA/dual-port hardware and RAID
- High power/TB
- System Memory (DRAM) requires many memcopies, resulting in limited performance
- Use of RAID means long rebuild times and limits capacity of SSDs
- Rebuild speeds on local RAID can take days. Limits capacity of SSDs to sizes that can be rebalanced within 24 hours or less.

This architecture was intended for a single storage server with small capacity

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What Has Changed in the Marketplace Since Then

The concept of storage has evolved a lot since PCIe Gen4 servers

New Driving Factors	Notes
More performance • Higher BW/TB • Higher IOPS/TB	As the capacity of drives has increased, the relative performance (per TB) of the drives decreases.
Higher concurrency	As files begin to server more purposes, the number of clients that may access the same files increases
Much higher capacity (> 1PB)	As file storage becomes more shared, there is increasing need for it to become centralized. Rather than store individual files across thousands of laptops, it needs to be network accessible in a centralized location (e.g. Sharepoint)
AI	Of course

Changing requirements requires new architecture

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AI Changes the Formula

AI storage requirements vary a lot from a legacy storage server

Factor	Normal Enterprise Use	AI Inference
Performance needed	Moderate (10s GB/s)	Very High (> 1TB/s) Requires clustered solutions
Capacity needed	100s TB	> 1PB
Concurrency needed	Moderate	Very high
Uptime	Very High	Varies, depending on specific use case, but generally high
Resiliency	Extremely high	Low (AI systems are rarely gold copies of data. Data may be a cache and not need any resiliency)

The differences in requirements forces new architectures for AI Inference Storage Servers

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AI SSD Use Cases: The Situation in 2024

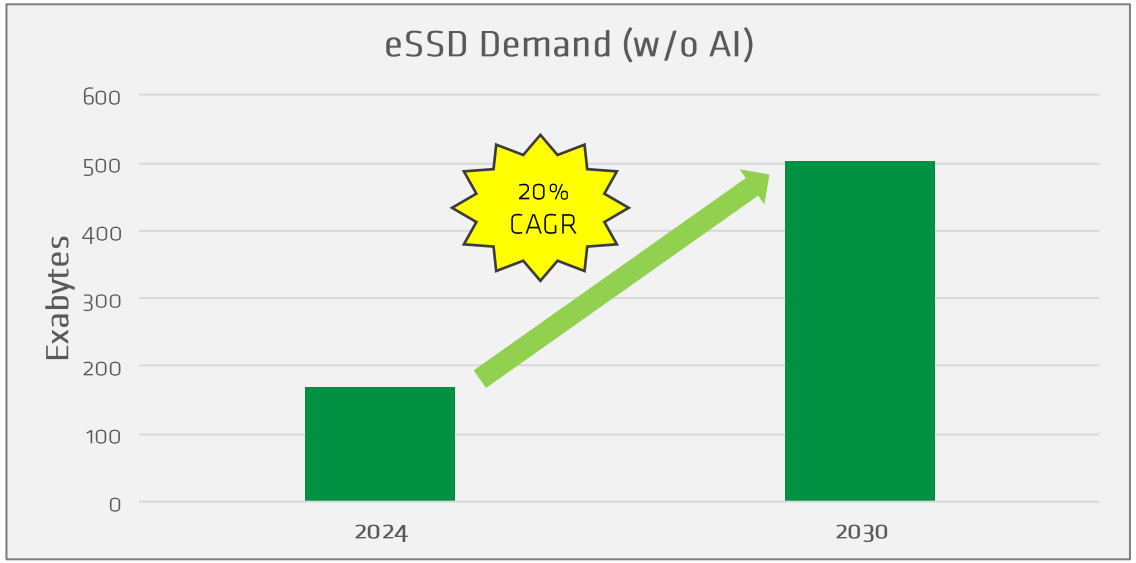
A simple question asked in 2024

- Question in 2024:

  How will AI impact the storage market for SSDs?

- Answer in 2024:

  Very little



From: https://www.mckinsey.com/industries/semiconductors/our-insights/generative-ai-spurs-new-demand-for-enterprise-ssds

The reasons:

- AI training workloads are **mainly GPU/network bound**. Faster storage will not help training
- Most AI training workloads can be handled by traditional storage solutions
- HDD performance on well tuned storage system was sufficient for training some models
- The need for SSD storage for AI inference was low

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AI SSD Use Cases: The Situation in 2026

Have things changed in 2 years?

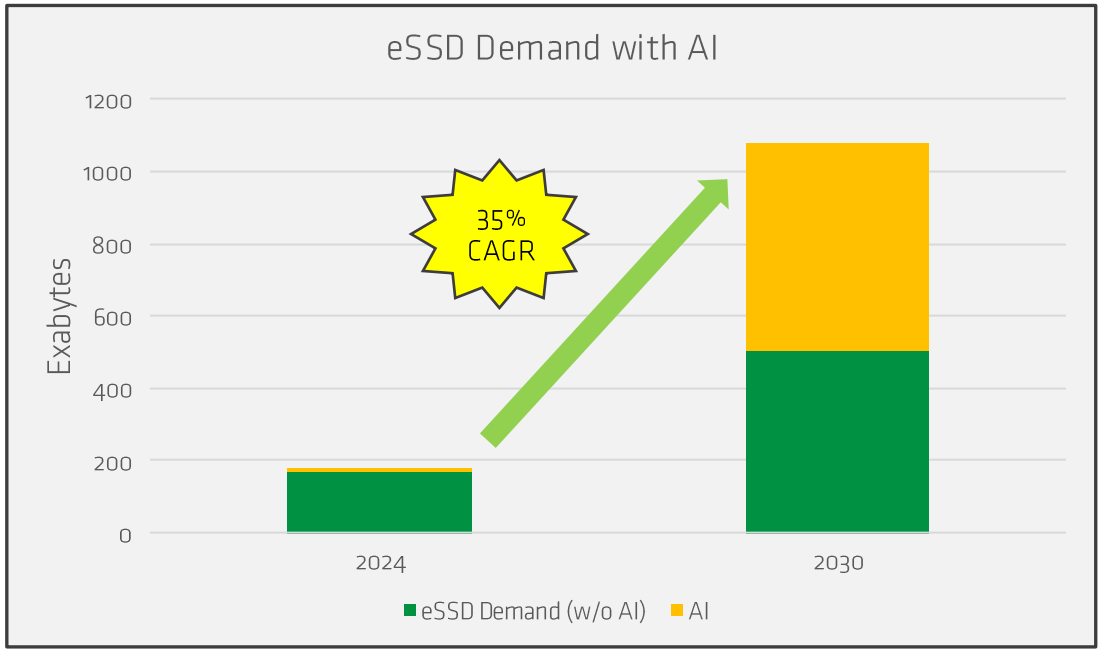
- Question in 2026:

  How will AI impact the storage market for SSDs?

- Answer in 2024:

  Huge changes. SSDs are essential for AI business overall.

  Performance is significantly better than HDD, but costs are much better than DRAM



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The reasons:

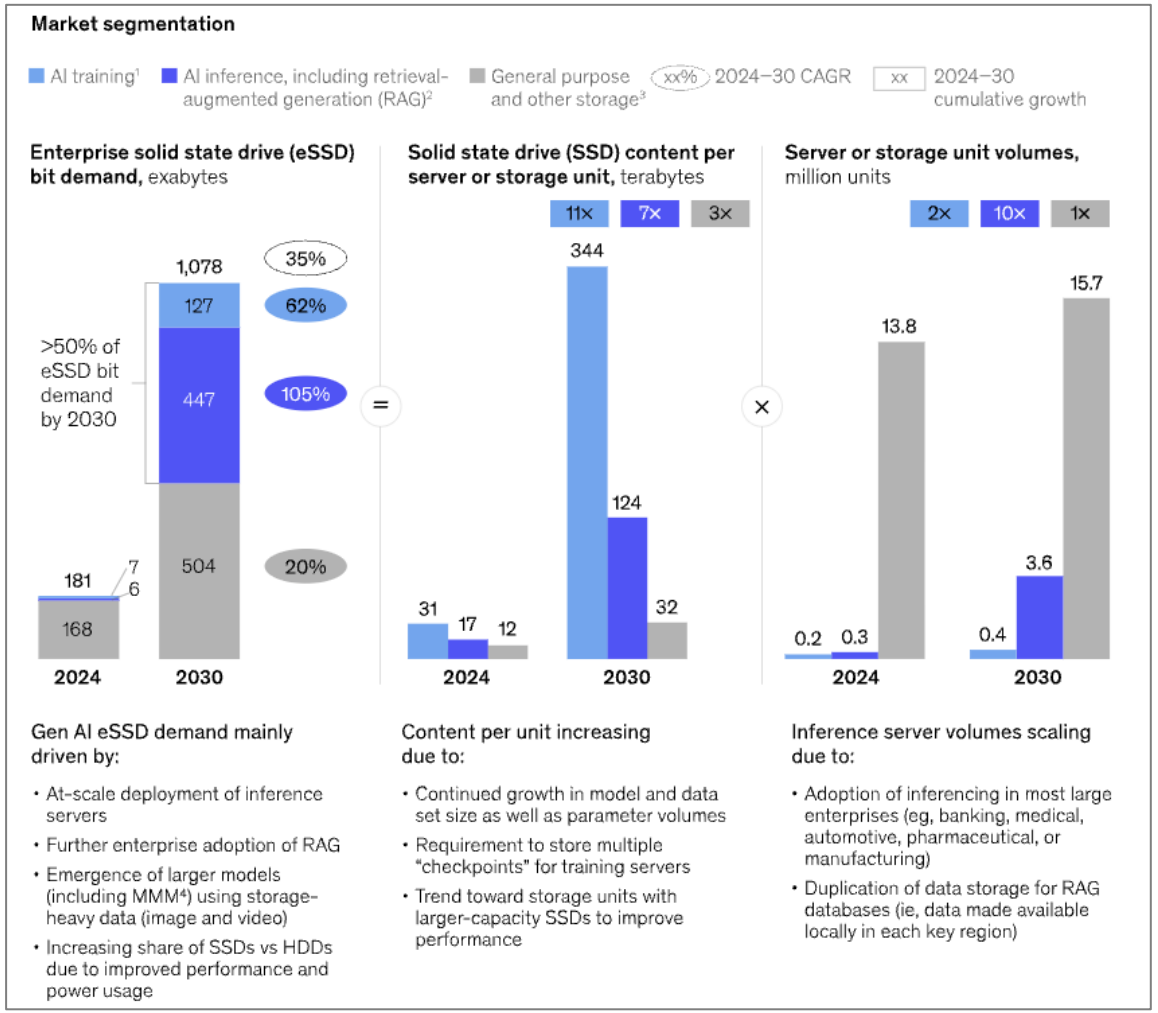
- Most of the focus is no longer on AI training, but on AI inference
- RAG solutions have been developed that require SSDs to store data for inference
- Training times of up to 1 month mean that checkpointing is very important. These require very high bandwidth read/write performanc e which requires SSDs (compared to HDD)
- Biggest reason: AI inference of LLMs require multiple prompts per session. Storing the contexts of these prompts on SSDs is crucia l for increasing GPU utilization. Nvidia solution is called KV Cache. Without context storage, GPU utilization may be only 10-20%. 60 % is seen as the level needed for AI to be economically viable.

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AI Storage: What the Market Looks Like

Key Points

- Primary market driver is deployment of AI inference, with much of the capacity being taken by context storage (KV Cache)
- SSD content per server is expected to multiply as models grow larger and more complex
- Models will double in size every 3 years
    - Plateaus in 2030
- Total eSSD market will grow 35% annually from 181 EB in 2024 to 1,078 EB in 2030
- 447 EB of AI inference. @$0.5/GB, this is more than $200B, just for AI inference



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Definition of AI Inference Server

Now let's define what an AI Context Storage is:

An AI Context Storage Server is built to serve up files/blocks/objects to clients. It has the following properties:

- Contains compute to handle requests. May be f/b/o or a combination.
- Contains many SSDs for base storage .
- As a cache, it does not need to provide resiliency. This maximizes performance and reduces overall TCO costs.
- Typically intended to handle total volumes of > 1 PB

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Possible Future of Storage for AI Inference

How might an AI context cache server architecture change in PCIe Gen 6?

Blade Design
- A 1 or 2 U Chassis can support multiple independent blades
- HA can be handled by Erasure Coding rather than HW HA
- Lack of hardware HA can mean savings on SSD costs
- Smaller blades reduces blast radius
- Blades do not need to filled with SSDs at the start

400/800 Gb NIC (RoCE)
- Support for higher speed networking @ up to 800 Gbps
- Integrated networking minimizes power

1200+ W PSU
- A single DPU can use much less power than CPU + NIC + Switch

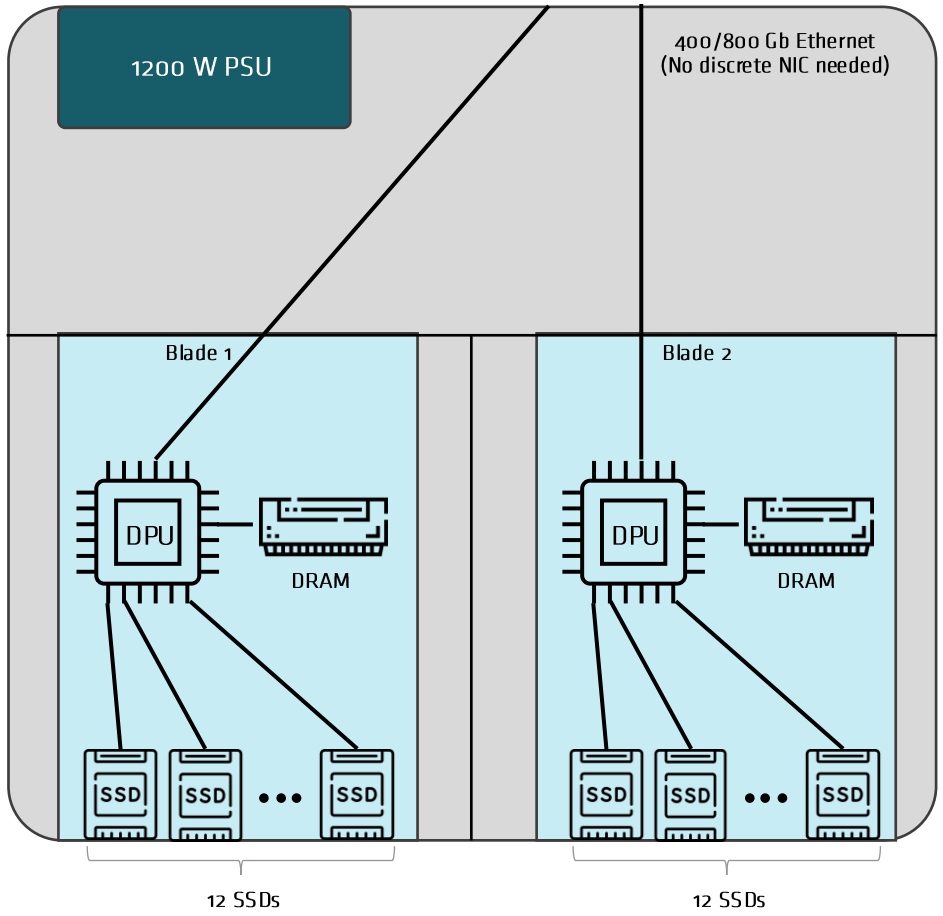
1 x 32 core DPU per module
- DPU incorporates lower power cores with NIC functions
- CPU bypass means smaller core counts are needed
- Power ~125W per DPU

64 GB DRAM
- Less DRAM needed since system memory will be bypassed

PCIe Switches
- Not needed

12 x 16+ TB SSDs (E3.s form factor)
- Higher capacity desired
- Moderate endurance needed (~1 DWPD)

Modular Bladed AI Inference Server



Issues

- Must have appropriate software architecture
- To make use of Erasure Coding, must start with a large number of storage nodes (this is optional for Context Caching anyway)
- Lack of major DPU server support (other than Nvidia)
  - These are still new

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Recommendations for AI Inference

1. Reconsider all legacy rules of thumb!!!
2. Look for solutions that minimize the software/hardware stack between XPU and media device
   - Uses NVMe natively (avoid SATA/SAS)
   - RDMA (RoCE or Infiniband)
   - Uses pNFS or similar technology to make parallel requests
3. Be careful with rack power limits when investigating servers. Storage servers can take up to 3,000 Watts of power. A normal rack can supply less than 14,000 Watts, so you may only be able to put 4 servers into a rack.
4. Because context caching can be recalculated, avoid added resiliency methods (RAID or Erasure Coding). The cost of the added resiliency in capacity is small, but it will hurt performance dramatically.
5. Look for higher capacity SSDs coming soon. These will help to reduce TCO by minimizing non-SSD costs, reducing OPEX, and saving space. TCO calculations need to be based on cost/usable GB/year.

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